Farès Zaghouane

Machine Learning Research Engineer · MSc EPFL

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SUMMARY

Machine learning research engineer (MSc EPFL, 2026) working across deep learning, computer vision, reinforcement learning and formal verification. At Nokia Bell Labs, took an open problem with no dataset and no agreed metric to a working video pipeline for cloud gaming; at Apple, built tooling that turns specifications into formal checks for hardware. Looking for research engineer and ML engineer roles in AI research teams.

WHAT DRIVES ME

Machine learning for physical systems that evolve over time and must be controlled or checked, built on computer science and electronics, hardware verification at Apple and research on world-model dynamics. Aim: robotics and automation.

EXPERIENCE

Nokia Bell Labs — Stuttgart, Germany

Research Intern (MSc Thesis), focus-map prediction for cloud gaming — *Aug 2025 – Feb 2026*

- Predicted which parts of each video frame matter to the player, so a cloud-gaming encoder can spend its bandwidth there. Team of three: two supervisors and me.
- Designed with my supervisor a metric that allows training without hand-labeled data; built the data engine alone: about 2k semi-manually annotated keyframes are used to fine-tune a VLM that drives automatic labeling of about 2 h of gameplay from 12 games across 4 genres.
- Built the pipeline (Grounding DINO, RAFT optical flow, lightweight correction network): the heavy semantic model runs on one frame in eight, an estimated 5× cost cut versus running it on every frame; beats the motion-only and saliency-only baselines. Compared it with a JEPA-style predictor and state-space models under latency and compute limits.

Apple — Munich, Germany

Design Verification Research Intern, formal verification of hardware — *Oct 2022 – Jul 2023*

- Built a Python framework that turns finite-state-machine specifications into Verilog/SystemVerilog assertions for a commercial model checker, so checks are written above the level of individual clock cycles.
- Evaluated it on industrial designs already verified by hand, with bugs injected on purpose. Research project in a 30-person team, carried out mainly by me with one engineer supervising.

PROJECTS

RL from Scratch

25 reinforcement-learning algorithms rebuilt in PyTorch on one shared runtime, from Q-learning to PPO/SAC, Dreamer, MuZero and an action-conditioned JEPA world model. HalfCheetah-v5: SAC/TD3 ≈10k, DDPG ≈11k near 1M steps.

Machine Learning from Scratch

52 course notebooks and a 123-module scikit-learn-style library, from a hand-written autograd engine through Transformers, state-space models and diffusion to optimal transport.

Structured Latent Dynamics *(research, in progress)*

Does an explicit dynamical structure make a world model's long-horizon predictions more stable? Controlled comparison against MLP, transformer and selective-SSM predictors, with paired runs and an oracle-basis control.

Constrained Trace EBM *(research, in progress)*

Energy-based model that scores execution traces against formal specifications, evaluated on unseen machines against an exact symbolic checker.

GPT-2 from Scratch

GPT-2-style language model written from the original papers: tokenization to pretraining, generation and fine-tuning.

EDUCATION

École Polytechnique Fédérale de Lausanne (EPFL) — Lausanne, Switzerland

MSc Computer Science, Machine Learning specialization — *Sept 2023 – 2026*

CPE Lyon — Lyon, France

Diplôme d'Ingénieur, Computer Science (Software Engineering & Big Data) — *Sept 2021 – 2026*

SKILLS

Programming & tools: Python, PyTorch, NumPy, TensorFlow, Git, Linux

AI & machine learning: Deep learning, computer vision, video understanding, self-supervised learning, Transformers, diffusion models, reinforcement learning

Models & systems: World models (JEPA), state-space models, dynamical systems, energy-based models, VLM fine-tuning

Verification & method: Formal verification, model checking, FSMs, SystemVerilog assertions, controlled experiments

Languages: French (native), English (C1, Cambridge CAE)